

MODEL QUESTION PAPER

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

NON RELATIONAL DATABASE MANAGEMENT SYSTEMS-SET-I

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. What is No-sql database?
2. What do you mean by key-value data model?.
3. What is a Btree?
4. Define hashing.
5. List any two indexing methods.
6. Define atomicity.
7. Define data integrity.
8. What is database security?
9. List two types of locks.

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Mention different types of no sql datatypes
2. Briefly describe graph-based database.
3. Define hash rings and its use.
4. Difference between dense index and sparse index.
5. What is a B+ tree. Explain its structures.
6. Explain properties of transaction.
7. Describe precedence graph.
8. Differentiate between strict two phase locking protocol and rigorous two phase locking protocol.

9. What is a named system exception in Oracle?
10. Mention any three pl sql security threats.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Explain in detail about document data model and key value data model

OR

2. How NoSQL system handle big data problem.

3. Explain dynamic hashing.

OR

4. Explain Btree insertion with an example.
5. Briefly explain different indexing in dbms.

OR

6. Explain different file organization and file operation methods.
7. Explain two phase locking method.

OR

8. Explain any two problems with concurrent execution.
9. Briefly explain about transaction states.

OR

10. Explain the concept of recoverability in dbms.
11. Explain briefly about database security threats.

OR

12. Explain about pl/sql locks in dbms.

MODEL QUESTION PAPER
FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
NON RELATIONAL DATABASE MANAGEMENT SYSTEMS-SET-II

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. Define hash ring?
2. The queries should be moved to the data rather than moving data to queries. Is it true?.
3. What is a B+ tree?
4. Define dynamic hashing.
5. List any two file organization technique.
6. Define transaction.
7. Define consistency.
8. Define lock.
9. List two defence mechanism in dbms.

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. What are the key features of document database?
2. What is query processing in distributed database?
3. Briefly explain columnar data model.
4. Explain Btree deletion.
5. Explain different types of file organization.
6. Check the view serializability for the schedules W3(X), R2(X), W2(Y), R1(Z), W3(Y), W1(Y).
7. Draw the states of transaction.
8. Mention three levels of locking

9. Mention any three security and integrity threats
10. Explain briefly about statistical database auditing.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Explain in detail about graph based data model.

OR

2. Explain briefly about distributed database system.
3. Explain primary indexing.

OR

4. Explain B+ tree with an example.
5. Briefly explain hash based indexing in dbms.

OR

6. Explain different file organization and file operation methods.
7. Explain acid properties in transaction

OR

8. Explain scheduling in dbms.
9. Explain the concept of recoverability in dbms.

OR

10. Explain the concept of indexing in dbms.
11. What is database security and why is it important?

OR

12. Explain about explicit locks in dbms.

(6x7=42)